

Phase III: Software Design and Modeling (TinyTrackers)

Submission Deadline: **04/05/2026 – 23:59**

In Phase III of the project, you will focus on software design and modeling. Please follow the instructions below.

Note: Examples below are used referring to a Cinema/Movie Reservation System.

1. Software Architecture

Overall Structure

The system follows a **three-tier architecture**:

Frontend (Presentation Layer) → used by Parents, Staff, Admin

Backend (Application Layer) → handles logic, authentication, notifications

Database (Data Layer) → stores all system data

How the system works

Scenario: Parent views child activity

Parent opens dashboard (Frontend)

Parent selects **child profile**

Request goes to **Backend**

Backend checks **Database**:

Child information (name, age, parent_id)

Activities (meals, sleep, behavior)

Photos

Pickup records

Backend processes data

Backend sends results back to **Frontend**

Parent sees **activities, photos, updates**

Scenario: Staff updates activity

Staff logs activity (Frontend)

Request goes to **Backend**

Backend validates input
Backend stores data in **Database**:
 Activities table (type, description, timestamp)
Backend triggers **Notification Service**
Parent receives **real-time notification**

Scenario: Child pickup

Staff marks child as **picked up**
Pickup person enters **safe word**
Backend verifies safe word from **Database**
If correct:
 Pickup stored in **Pickups table**
 Notification sent to Parent
If incorrect:
 Access denied

Component Diagram (UML)

Provide a diagram showing the main components of your system and how they interact.

Quantity: 1 UML diagram

1.Top Layer → Users (Actors)

Parent / Staff / Admin

These are the **stakeholders**:

Parents → View activities, photos, notifications

Staff → Update child data (meals, sleep, behavior)

Admin → Manage users and system

2. Frontend (Presentation Layer)

What it does:

Displays dashboards (Parent / Staff / Admin)

Sends requests to backend

Shows results (activities, photos, alerts)

Examples:

Parent clicks “View Activities”

Staff clicks “Add Meal”

3. Backend (Core Logic Layer)

What it does:

Processes all requests

Applies business logic

Validates data

Connects all services

4. Core Components (Middle Section)

These are the main functional modules

User Management

Handles login, registration

Manages roles (parent, staff, admin)

Child Management

Stores child data

Links children to parents

Activity Service

Records:

Meals

Sleep

Behavior

Photo Service

Uploads images

Displays photos to parents

Pickup Service

- Handles child pickup process

- Stores pickup records

SafeWord Verification

- Checks if safe word is correct

- Prevents unauthorized pickup

Notification Service

- Sends:

 - Activity alerts

 - Pickup alerts

 - Real-time updates

Reports Service

- Generates history reports

- Shows past activities

How these components interact

Example flow:

- Activity Service → triggers → Notification Service

- Pickup Service → uses → SafeWord Verification

- All services → use → Data Access Layer

5. Data Access Layer

What it does:

- Connects backend to database

- Performs:

 - INSERT

 - UPDATE

 - DELETE

SELECT

Example:

Save activity

Fetch child data

Database (Data Layer)

Stores everything:

Users

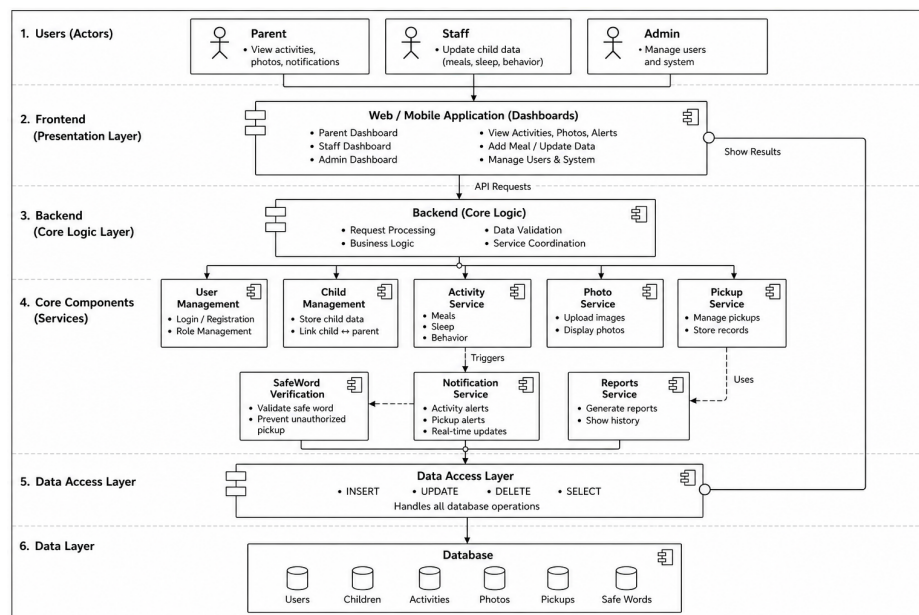
Children

Activities

Photos

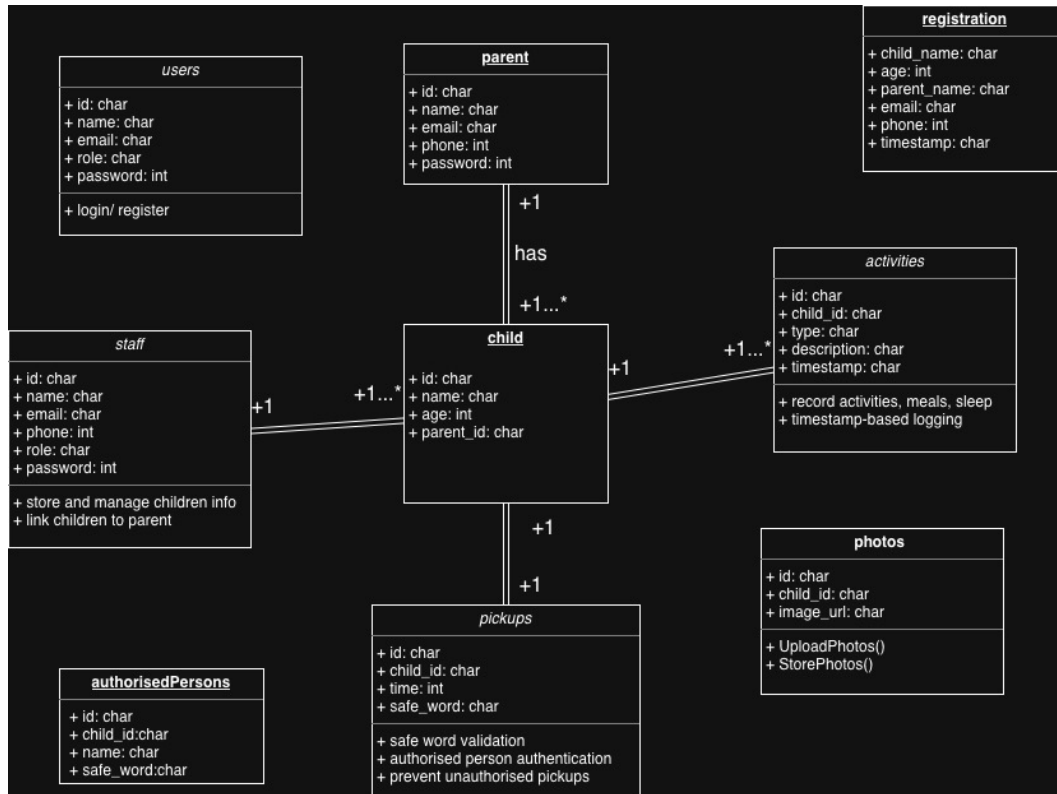
Pickups

Safe words

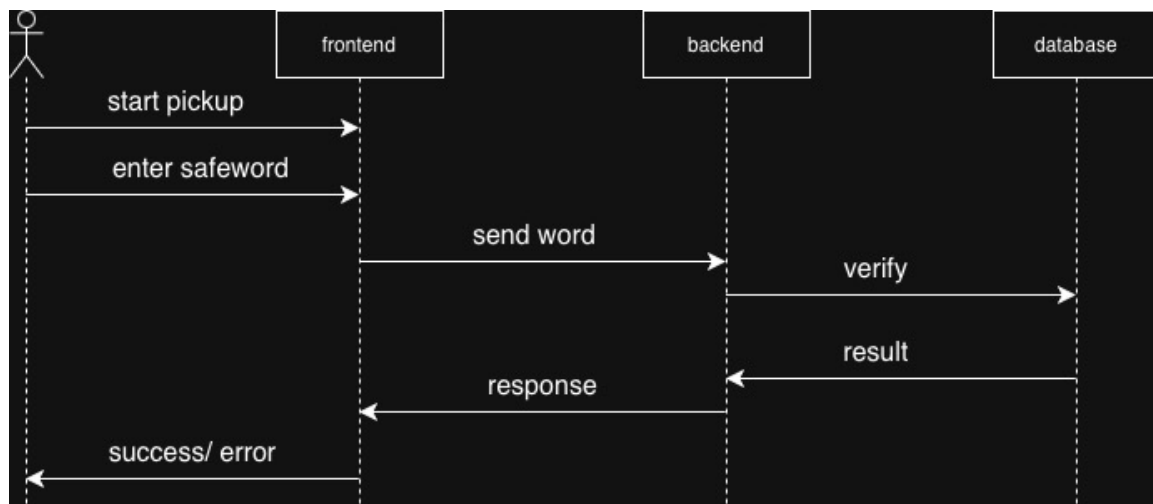


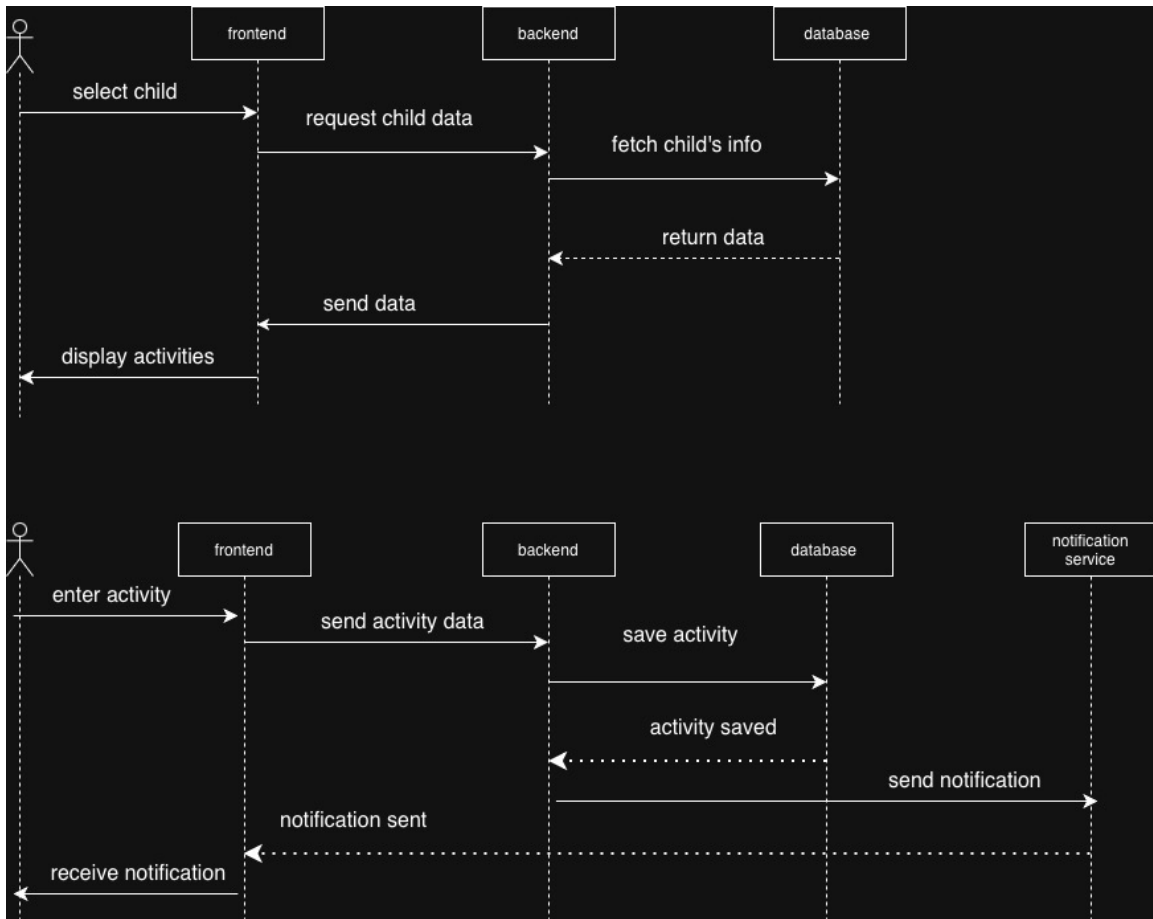
2. Detailed Design

Class Diagram



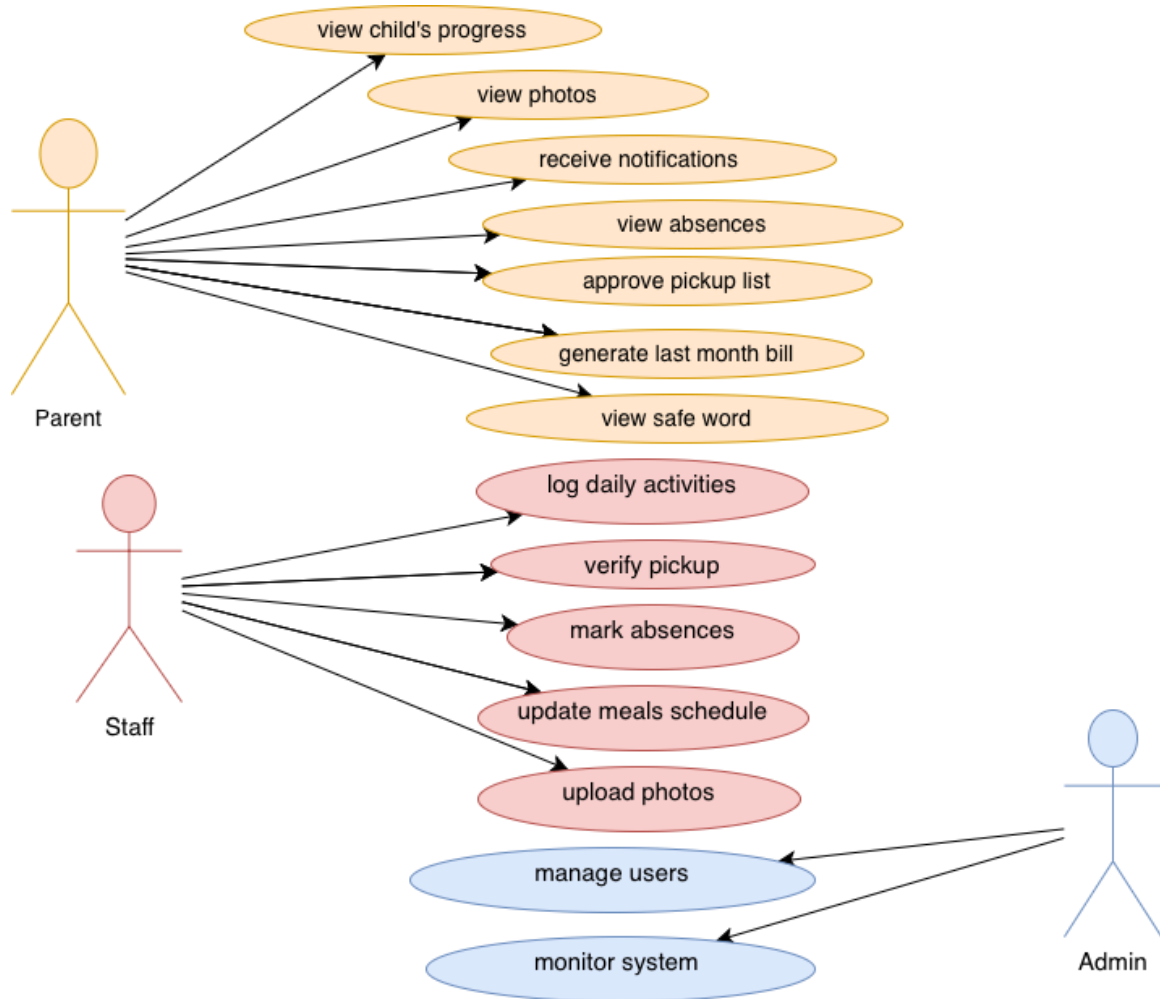
Sequence Diagrams





3. Modeling

Use Case Diagram



Activity Diagrams



State Diagram

